Christopher Egly

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EDUCATION

University of Illinois at Urbana-Champaign, Grainger College of Engineering — **Expected May 2027**
Bachelor of Science in Mechanical Engineering, Focus on Robotics and Controls — GPA: 3.97/4.0

EXPERIENCE

Intern at SpaceX *Payload Integration Intern* — August 2026 – December 2026

Honeybee Robotics/Blue Origin (*due to NDA, some bullets are vague*) — Altadena, California
Mechanical Engineering Intern — May 2026 – August 2026

- Delivered critical path testing results for the MK1 VIPER Offloader, utilizing a custom test rig designed and built on testing criteria collected from relevant stakeholders to meet customer requirements
- Created and documented a standardized process for assembling flight tether "end terminations"
- Validated on custom test rig in extreme environmental conditions (-125C)
- Discovered, investigated, and elevated critical design oversights to relevant team members demonstrating mastery of "weakest-link" design philosophy

Stryker Sustainability Solutions — Phoenix, Arizona
Manufacturing Engineering Intern — May 2025 – August 2025

- Reverse engineered, and documented a SQL database with over 250 tables to implement a Power BI system health monitoring dashboard for facilities, quality and manufacturing teams to track for preemptive downtime maintenance, avoiding future unnoticed incidents at a cost savings of $274k in absorption from one unscheduled down
- Designed a fixture, PCB, and process to harvest connectors from used medical devices for testing reprocessed EP catheter cables, cutting lead time by 10 weeks and downtime by 2 days per failure

Argonne National Lab, DOE SULI program — Lemont, Illinois
Mechanical Engineering Intern — May 2024 – August 2024

- Developed and tested a closed loop, machine learning driven control system for a piezoelectric x-ray bimorph mirror by combining two NNs in series to predict future mirror profiles and adjustments. Link to finished project
- Improved mirror surface accuracy by 30% to less than 1.32 nanometers RMS and added key stabilization technique restraining mirror creep by 67% to within 0.081 nanometers RMS by utilizing tandem neural network techniques
- **Resultant publication:** R. Zhang, L. Rebuffi, C. Egly, et al., "Tandem neural network-based controller for x-ray bimorph mirrors", *Opt. Letters* 50, 10 (2025). DOI: 10.1364/OL.559583

PROJECT & RESEARCH HIGHLIGHTS

Novel Mobile Robotics Laboratory, Department of Mechanical Engineering, UIUC — Champaign, Illinois
Undergraduate Researcher — August 2025 – Present

- Refactored and merged code base from C to C++ oriented design, compile speed reduced to negligible level
- Tested hopping, rolling, and balancing capabilities at the SLOPE lab, NASA GLENN in Ohio

6-Axis Robot Arm, American Society of Mechanical Engineers (ASME) — Champaign, Illinois
Project Leader — September 2024 – April 2025

- Lead a group of 14 engineers designing a 6-DOF robotic arm loosely inspired by the Modern Robotics UR3 for presentation at the University of Illinois's Engineering Open House (EOH) event
- Utilized forward and inverse kinematic calculations in Python to define current and future states, convert between the world frame and configuration space, and create a visual simulation of different robot configurations

SKILLS

Design & Analysis: SolidWorks, Inventor, Fusion, Creo, Altium, Arduino (PLC), Simulink, GD&T, Soldering
CS: Embedded systems (TMS320F28379D), Python, C, C++, MATLAB, Pytorch, SQL, Excel, NumPy, Java, Power BI
Related Coursework: Mechatronics (SLAM and Autonomy), Robot Dynamics and Control, Design I, II & III